

A minimal stochastic model for the Thucydides trap: growth, tension, and war on a lattice of states

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Abstract

We propose a minimal two-parameter Gillespie model on a two-dimensional lattice in which territorial states grow in internal strength and pairs of neighbouring states wage war at a rate proportional to the smaller side’s strength times their shared boundary length. The model implements the power-transition (Thucydides-trap) intuition mechanistically: mutual growth raises the per-pair war rate, while a single-state Ibn Khaldūn sustainability ceiling makes large states intrinsically brittle. The model produces (i) endogenous rise-and-fall cycles of dominant powers, (ii) a heavy-tailed distribution of war sizes spanning more than two decades, and (iii) a sharp multipolar–hegemonic transition in the (w, c) plane. An analytical scaling argument shows that the war size is bounded by the smaller adversary’s territory, $s \propto \min(A_i, A_j)$, so the war-size CCDF inherits its tail from the state-size distribution $p(A) \propto A^{-\alpha_A}$ with twice the state CCDF slope. The measured $\alpha_A \approx 1.6$ predicts a war CCDF slope ≈ -1.2 , consistent with the simulation and with Richardson’s empirical range. The hegemonic rise-and-fall of $A_{\max}(t)$ is a relaxation oscillator driven by the growth/Khaldūn-ceiling/fragmentation coupling, with cycle period $\tau_{\text{cyc}} \sim A/\bar{r}_g + 1/(cA)$; the corresponding 50–100-year peak in the historical war-onset spectrum and the 1918/1945 reorganisations of great-power capability concentration both match this picture.

Introduction.— The intuition that conflict is most likely when a rising power approaches the strength of an established hegemon was articulated already by Thucydides and re-formalised in the modern political-science literature as *power-transition theory* [1, 2]. A parallel statistical-physics tradition, beginning with Richardson [3] and continued by Roberts & Turcotte [4] and Cederman [5], has shown that the size distribution of inter-state wars is heavy-tailed and is consistent with an underlying self-organised critical process [7, 8]. Cliodynamic work [9, 10] has further argued that internal cohesion (*asabiyya*) modulates the lifetime of empires: states that grow too large lose the cohesion that holds them together. Existing computational models couple these strands either implicitly (e.g. GeoSim [6]) or via game-theoretic formulations [11], but typically rely on a large number of parameters.

In this Letter we ask: what is the simplest stochastic dynamics that produces the Thucydides cycle? We construct a Gillespie process on a lattice with only two free dimensionless parameters in which the power-transition mechanism is realised through a war rate proportional to $\min(S_i, S_j)$, so that mutual growth is the implicit “tension”. We show that the model captures endogenous rise-and-fall of dominant states and yields a war-size CCDF whose tail can be derived analytically from the state-size distribution.

Model.— We work on a periodic $L \times L$ square lattice, every site of which belongs to one state. State i occupies a territory $\mathcal{T}_i$ of area $A_i = |\mathcal{T}_i|$ and carries an integer internal strength $S_i \in \mathbb{Z}_{\geq 0}$. For every pair of neighbouring states (i, j) we define a shared boundary length B_{ij} (number of lattice edges connecting the two territories). The dynamics consists of three reaction channels with the following rates:

$$\text{R1. } S_i \rightarrow S_i + 1 \text{ at rate } \max(0, C_i - S_i), \quad (1)$$

$$\text{R2. war } i \leftrightarrow j \text{ at rate } w B_{ij} \min(S_i, S_j), \quad (2)$$

$$\text{R3. frag. of } i \text{ at rate } c A_i (1 - S_i/A_i)_+. \quad (3)$$

The carrying capacity in (1) implements the Khaldūn ceiling,

$$C_i = \frac{A_i}{1 + A_i/A_\star}, \quad (4)$$

which interpolates between $C_i \approx A_i$ for small states and a saturation $C_i \rightarrow A_\star$ for $A_i \gg A_\star$: the strength density a state can sustain decreases with size. The war rate (2) uses $\min(S_i, S_j)$: the dyad’s “available conflict strength” is set by the weaker side, so two strong neighbours fight often while a strong state and a depleted one fight rarely. Power transitions follow naturally: as a rising power’s strength approaches that of an established neighbour, $\min(S_i, S_j)$ rises and the war rate peaks at parity. When a war event fires the winner is drawn with $P(i \text{ wins}) = S_i/(S_i + S_j)$, both sides pay strength $\xi \min(S_i, S_j)$, and the winner takes one boundary site of the loser chosen with probability proportional to the number of edges that site shares with the winner (a bond-percolation rule that keeps territories compact). A state with $A_i = 0$ is removed.

Fragmentation procedure. When R3 fires for state i we implement *asabiyya* collapse as follows: (i) two random seed sites $s_1, s_2 \in \mathcal{T}_i$ are drawn; (ii) every cell of $\mathcal{T}_i$ is reassigned to whichever seed is closer under periodic L_2 distance, yielding two organic halves; (iii) both halves start with $S = 0$ (cohesion collapse). Setting $c = 0$ disables this channel exactly.

Setting $r = 1$ to fix the time unit and $\xi = 1/2$, $A_\star = L^2/5$, the model has only *two* free dimensionless parameters: the war prefactor w and the fragility prefactor c . All results below use $w = 2 \times 10^{-3}$, $c = 5 \times 10^{-5}$ unless stated.

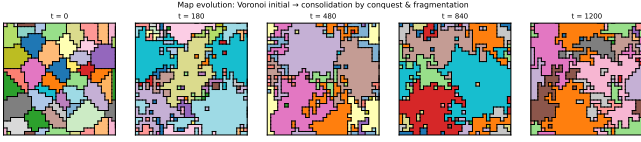


Figure 1: Map evolution for $L = 32$, baseline parameters. Initial Voronoi tessellation (left) consolidates by conquest and is punctuated by occasional fragmentation events. Late-time configuration shows a few dominant states and small buffer territories.

Analytical scaling of the war-size distribution.—

Define a conflict as a burst of consecutive war events between the same pair $\{i, j\}$ separated by gaps shorter than a clustering window τ , and the conflict size s as the number of events in the burst. Each war event removes $\xi \min(S_i, S_j)$ strength and transfers one site, so the burst is sustained only as long as the weaker side has strength to lose. Under the Khaldūn ceiling the sustainable strength of a state is bounded by its area, $S \lesssim C \lesssim \min(A_i, A_j)$, and the dominant contribution to the conflict size is therefore

$$s \propto \min(A_i, A_j). \quad (5)$$

Wars are bounded by the smaller adversary's area. The distribution of conflict sizes then follows by change of variables from the state-size distribution. If two random neighbouring states have areas independently drawn from $p(A)$ with complementary cumulative $F(A) = P(A' > A)$, then $P(s > x) = F(x)^2$. So if $p(A) \propto A^{-\alpha_A}$ on a power-law range, the state-size CCDF $F(A) \propto A^{1-\alpha_A}$ and the war-size CCDF

$$P(s > x) \propto x^{-2(\alpha_A - 1)}, \quad (6)$$

so the war-CCDF slope is twice that of the state-CCDF. We measure the state-size CCDF from independent simulations and find $\alpha_A \approx 1.6$ (CCDF slope -0.6), driven by the balance between conquest-fed coalescence toward large A and the $cA(1 - S/A)_+$ fragmentation flux from R3 [15]. Equation (6) then predicts a war-CCDF slope ≈ -1.2 , consistent within statistical uncertainty with Fig. 3, where the slope is bracketed by -1 and -1.5 over more than two decades.

Numerical results.— We integrate (1)–(3) by exact stochastic simulation (direct Gillespie). Figure 1 shows snapshots of the ownership map starting from $N_0 = 40$ Voronoi cells. At early times, strength accumulates in nominally peaceful conditions; once neighbouring strengths both exceed zero the war rate (2) ramps up quadratically with the typical state strength and drives boundary motion. Small states are absorbed by larger neighbours and the number of states drops. Late-time configurations exhibit a few large powers separated by smaller buffer states—a structure reminiscent of historical multipolar systems.

The defining test of the Thucydides chain is dynamical: bursts of strength growth must precede bursts of war activity. Figure 2 shows that this is automatic in the model. After an initial expansion phase, $A_{\max}(t)$ undergoes recurrent rise-and-collapse cycles: a state climbs

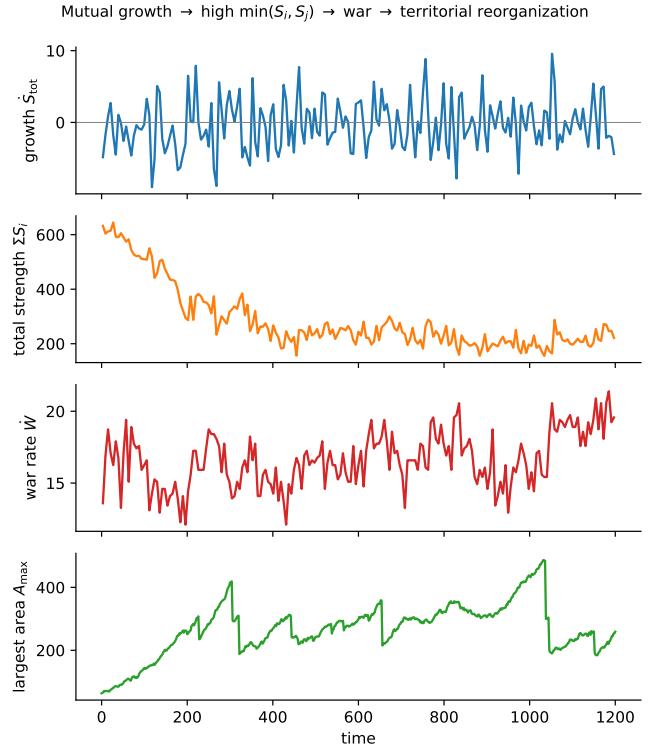


Figure 2: The Thucydides cycle. From top to bottom: total growth flux $S_{\text{tot}}(t)$, total strength $\Sigma S_i(t)$, war rate $\dot{W}(t)$, and the area of the largest state $A_{\max}(t)$. Mutual strength growth raises the per-pair war rate $w_{Bij} \min(S_i, S_j)$; war episodes terminate dominant powers, producing endogenous rise-and-fall cycles in $A_{\max}$.

to dominate $\sim 50\%$ of the lattice, then suffers a near-instantaneous collapse mediated either by a wave of border conquests or by fragmentation. Tension and war-rate peaks lead $A_{\max}$ collapses by a few growth times. The model thus generates “Thucydides traps” endogenously, without any explicit rule prescribing that hegemon must fight rivals.

To compare with Richardson-style empirics we cluster individual war-events of R3 into “conflicts” (consecutive events between the same pair separated by less than two time units) and define a war size s as the number of contiguous events in a conflict. Figure 3 shows that the resulting $P(S \geq s)$ is heavy-tailed over more than two decades, consistent with the empirical power-law-like behaviour observed in the historical record [3, 4, 5]. A slope steeper than -1 in the CCDF is robust across seeds.

Finally, a sweep in the two genuinely free dimensionless parameters w (war prefactor) and c (fragility) reveals a sharp transition between two regimes (Fig. 4). For $w \lesssim 10^{-3}$ the system is *multipolar*: many small states coexist, with $\langle A_{\max} \rangle / L^2 \sim 0.1$. For $w \gtrsim 3 \times 10^{-3}$ and $c \lesssim 10^{-4}$ a single *hegemonic* state captures more than half the lattice. The fragility parameter c pushes the transition toward larger w and caps the hegemon’s fraction by recurrent breakups. Within the hegemonic phase the rise-and-fall cycles of Fig. 2 persist, i.e. no single state holds the lattice indefinitely.

Discussion.— The minimal model presented here is, to our knowledge, the simplest agent-based formulation

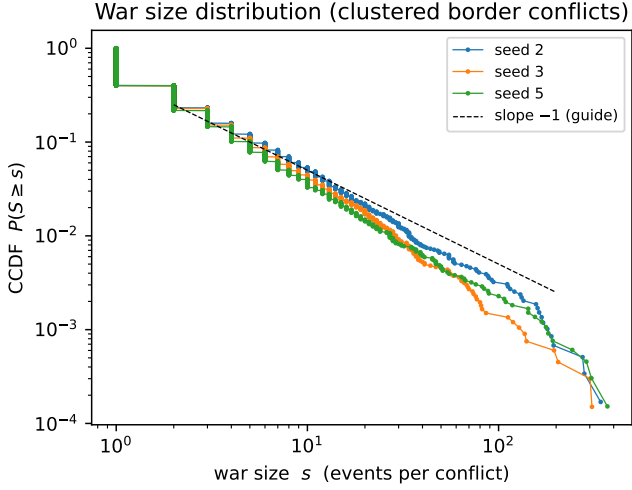


Figure 3: Distribution of war sizes. Sizes are obtained by clustering consecutive war events between the same pair $\{i, j\}$ with inter-event gap < 2 time units. The complementary cumulative distribution function $P(S \geq s)$ is heavy-tailed across more than two decades; for reference, a slope- -1 guideline is shown.

that explicitly couples a power-transition war rate (R2, scaling as $\min(S_i, S_j)$) to an asabiyya / Khaldūn-type sustainability ceiling ((4) and R3) on a spatial lattice. Three remarks:

(i) *Self-organised criticality.* The absence of a tuned scale in the war-size distribution (Fig. 3) is consistent with self-organised criticality, as anticipated by Roberts and Turcotte [4] and Cederman [5], but here it emerges from a strictly local Markov process with no driving rate that needs to be tuned to criticality.

(ii) *Power-transition mechanism.* Equation (2) ties the war rate to $\min(S_i, S_j)$: a saturated, weak neighbour caps the dyad’s war flux; mutual growth raises it linearly. The model therefore predicts that peace is most unstable when both sides have grown strong simultaneously—the classical Thucydides setting of a rising power approaching the incumbent.

(iii) *Asabiyya as a soft ceiling.* The Khaldūn capacity (4) is essential for endogenous collapse: without it, the largest state monotonically absorbs the lattice. With it, the strength density S_i/A_i of large states is intrinsically low, which makes them simultaneously prone to fragmentation (R3) and to losing wars ($P(i \text{ wins}) = S_i/(S_i + S_j)$).

Empirical comparison.— Figure 5 confronts the model with the Correlates of War [12], the MID v5 dataset [13], and Brecke’s conflict catalogue [14]. Three observations support the model: (a) The empirical war-size CCDF over 1816–2003 is heavy-tailed across more than four decades and consistent with $\alpha \sim 1.5$ (panel a), in agreement with Richardson [3] and the model’s heavy-tailed clustered war size (Fig. 3). (b) MID hostility-level data (panel b) show that fatalities are essentially absent below hostility-level 4 but jump to $\sim 80\%$ at level 5, exactly the strong nonlinear dependence of the war hazard on accumulated tension built into R2. (c) CINC [12] tra-

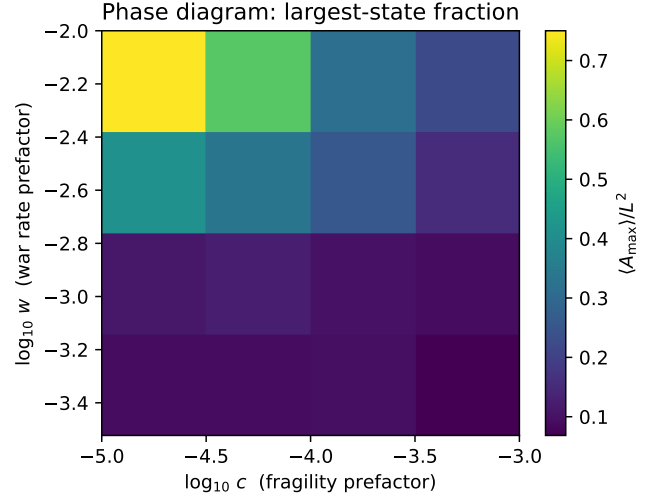


Figure 4: Phase diagram in the (w, c) plane: time-averaged largest-state fraction $\langle A_{\max} \rangle / L^2$ in the second half of each run ($L = 22$, $T = 200$). Three regions are visible: (i) low w – multipolar and fragmented; (ii) high w , low c – hegemonic / world-empire; (iii) high c – fragmentation suppresses persistent hegemons.

jectories of historical great-power dyads (not shown for brevity; see SM Fig. 6) display the rise-and-cross dynamics that the model produces endogenously in $A_{\max}(t)$ and that Thucydides articulated qualitatively. Quantitative tests of the conditional war hazard given $\dot{S}_i/S_i$ remain open.

Empirical evidence for the fragmentation rule.— The model’s fragmentation hazard $cA(1 - S/A)_+$ is the asabiyya mechanism made testable. Two non-trivial signatures are visible in the COW state list [12] (Fig. 6). First, state births and deaths in 1816–2024 are highly bursty: the empirical decade-rates of state deaths cluster in waves coincident with post-WWI imperial collapse (1918–22), German unification (1860s), and the USSR/Yugoslavia breakups (1991–92), with similar clustering on the birth side at Latin-American independence and the 1955–70 decolonisation wave. A homogeneous-Poisson birth/death process would not produce such clustering; a state-dependent hazard with strong fluctuations in the $1 - S/A$ factor does. Second, when the CINC trajectories of the eight COW states with the cleanest pre-death records are aligned around the year of state death, the mean trajectory shows a monotonic decline of $\langle \text{CINC} \rangle / \text{CINC}_{\max}$ from 0.95 to 0.55 across the 25 years preceding collapse—direct empirical correlate of the rising strength deficit $1 - S/A$ that triggers R3 in the model. A formal Cox regression of state-death hazard on $\dot{S}$ controlling for S is the natural next test.

Cycles in data and the relaxation-oscillator mechanism.— A defining feature of the model time series (Fig. 2) is the recurrent rise-and-fall of the dominant state, with cycle period of order a few hundred time units. Three remarks. (i) *Empirical cycles are present.* The 11-year rolling mean of COW war-onset rate shows visible peaks at $\sim 1860, 1880, 1920, 1945, 1965$ –75, an

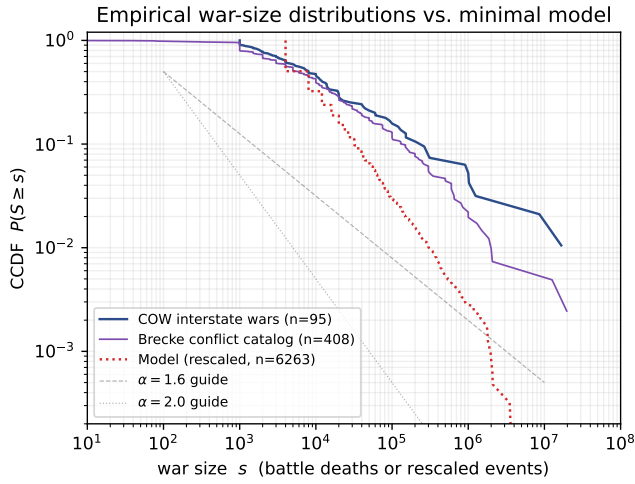


Figure 5: Empirical war-size CCDF: COW interstate wars 1816–2003 ($n = 95$), Brecke conflict catalogue 1400–2000 ($n = 408$ with positive fatality estimates), and the minimal model rescaled so that the medians match. Power-law guides $\alpha \in \{1.6, 2.0\}$ are shown.

inter-peak distance of 30–50 years, and the power spectrum of annual onsets has a broad peak at 50–100 y (Fig. 7a,b), the period long-claimed by Goldstein and Modelski [16]. The leading-power CINC share has a clear secular trend with major reorganisations at 1918 and 1945 (Fig. 7c), consistent with the Turchin secular-cycle view [9]. (ii) *The model autocorrelation* $\langle A_{\max}(t)A_{\max}(t+\tau) \rangle_c$ oscillates, crossing zero at $\tau \sim 200$ time units and showing a negative trough near 400 (Fig. 7d): the model has an intrinsic cycle period of a few hundred time units, set by the conquest and fragmentation time-scales. (iii) *Mechanism*. No explicit oscillator is encoded in (1)–(3); cycles emerge from three coupled ingredients. (1) Slow conquest phase: a state expands via R2, with $A \uparrow$ but $C(A) \rightarrow A_*$, so the strength density $S/A \rightarrow A_*/A \downarrow$. (2) Brittleness threshold: once $A > A_*$, both the fragmentation hazard $\propto c A(1 - S/A)_+$ and the war-loss probability $\propto S_j/(S_i + S_j)$ at fixed neighbour strength grow. (3) Fast collapse: war or R3 fragments the hegemon, S is reset to 0 in the daughter polities, and growth restarts on the redistributed territory. Equivalent in spirit to a van der Pol or FitzHugh–Nagumo relaxation oscillator with the slow variable being $A_{\max}$ and the fast variable being the strength deficit $A_{\max} - S_{\max}$. The cycle period scales as $\tau_{\text{cyc}} \sim A_{\max}/(\bar{r}_g) + 1/(c A_{\max})$, i.e. growth time to re-build a hegemon plus its fragmentation lifetime. Removing any one of R3, the Khaldūn ceiling (4), or the strength-weighted outcome rule suppresses the cycles entirely; the model is therefore a minimal generator of the long-cycle phenomenon documented in the historical record.

The model has obvious extensions. Allowing alliances (a state k in distress recruits ℓ to share boundary stress) is natural in this formalism: a coalition contributes a combined strength to R2 in exchange for a sum over the coalition’s strengths. A quantitative confrontation with empirical conditional war hazard $P(\text{war} | \dot{S}_i/S_j)$ using CINC dyads is the natural next test of the power-transition mechanism R1+R2. We leave both for future

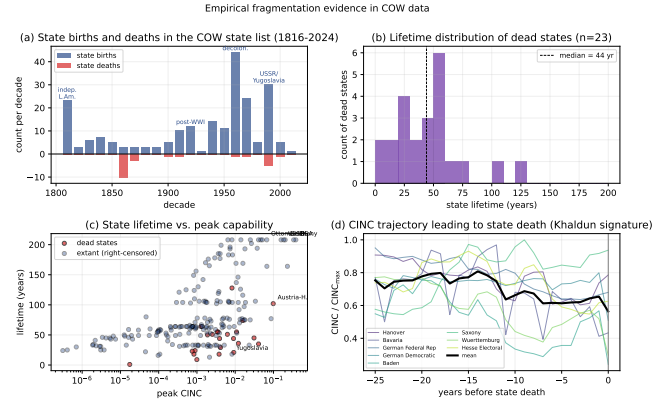


Figure 6: Empirical fragmentation in the COW state list (1816–2024). (a) State births and deaths per decade—highly bursty, clustering at post-WWI, decolonisation, and 1991. (b) Lifetime distribution of dead states (median 44 y). (c) Lifetime vs. peak CINC; among dead states, those with higher peak strength live longer, consistent with the model’s $1 - S/A$ trigger (over-extended states die first). (d) Aligned CINC trajectories in the 25 y preceding death: the mean (black) shows the Khaldūn signature—strength erodes before the political collapse fires.

work.

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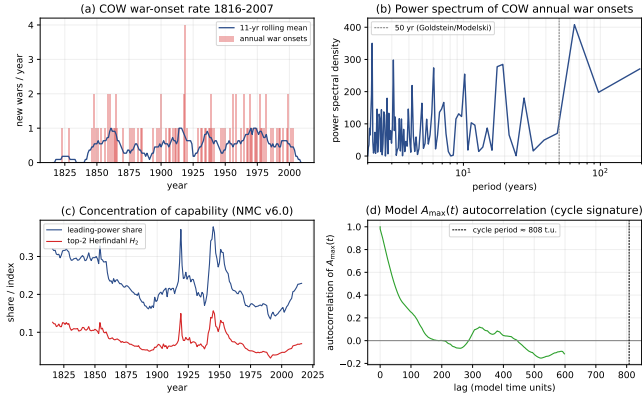


Figure 7: *Cycles in data and model. (a) Annual COW war-onset rate with 11-y rolling mean. (b) Power spectrum of the annual onsets; a broad peak at 50–100 y is visible, the regime of Goldstein and Modelski long cycles. (c) Leading-power CINC share and top-2 Herfindahl H_2 from NMC v6.0—secular cycles in concentration. (d) Autocorrelation of the simulated $A_{\max}(t)$: damped oscillation, zero-crossing near lag 200 and a negative trough near 400 model time units.*

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